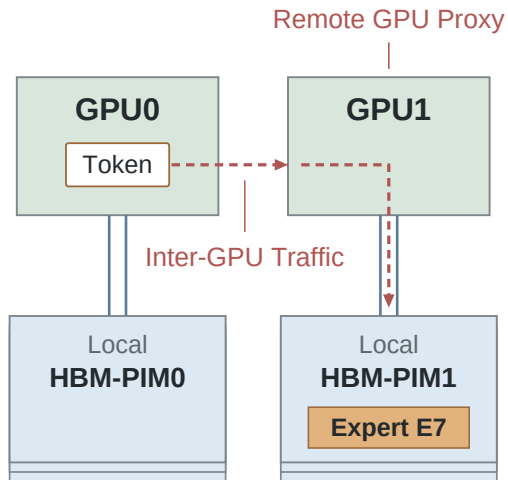


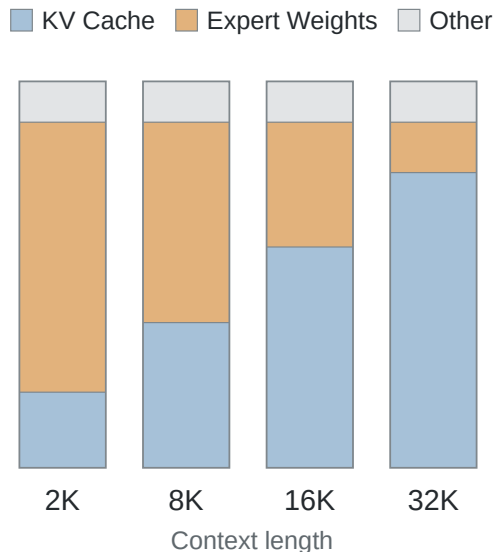
Motivation: Limitations of GPU-Local HBM-PIM for Multi-GPU MoE Inference

(a) Remote Expert Access



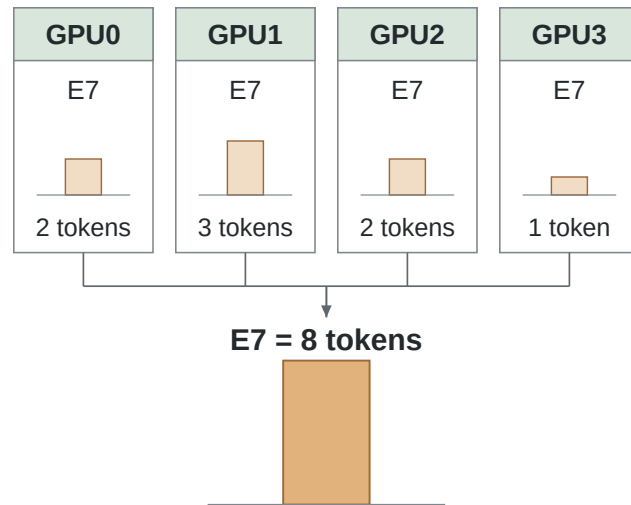
Extra Latency

(b) HBM Capacity Competition



KV Cache $\uparrow$ $\rightarrow$ Local Expert Capacity $\downarrow$

(c) Local vs Global Expert Hotness



Globally Hot

Locally Cold $\rightarrow$ Globally Hot

Local HBM-PIM

High Bandwidth · Low Latency · Locality

+

Missing Shared Capability

Large Capacity · Cross-GPU Sharing · Global Aggregation

Local and Shared PIM Domains Are Complementary